1. Corrected Table 1-style comparison

Existing checkpoints, matched baselines, no new training. This separate report uses the exact low-level "Retrained SAC" checkpoint from the PID-inclusive Table 2 comparison, plus its compatible original-bidding macro SAC partner. It does not replace the original manuscript or the prior PID comparison. Table 1A is a newly registered evaluation, not an exact replay of the archival producing runs.

Table 1A. Corrected, service-only results. Default scenario; `physical_v1`; start day 0; bias $\beta = 0$; seeds 100–104; 24 hours per case. Each cell gives the mean over all five cases, followed by ± sample SD ($n-1$ denominator). All SAC labels here mean the corrected checkpoints.

Macro + hardware	RMSE (kW)	Reg. rev. (USD)	Macro reward	Low reward	Warning fraction	Age (10^{-4})	Full 24h	All gates
Rule macro + no agent	2,619.5	4,611.6	−0.575	−0.341	0.000	0.00	5/5	5/5
	±185.4	±279.0	±0.024	±0.023	±0.000	±0.00		
Rule macro + Random	18,355.0	174.3	−3.906	−3.662	0.000	39.99	5/5	5/5
	±214.5	±41.2	±0.133	±0.133	±0.000	±0.11		
Rule macro + Bang-bang	998.7	5,450.0	−0.345	−0.119	0.000	55.34	5/5	5/5
	±239.0	±332.3	±0.059	±0.057	±0.000	±4.91		
Rule macro + Rule	604.7	5,493.5	−0.257	−0.031	0.000	27.40	5/5	5/5
	±263.4	±364.1	±0.034	±0.031	±0.000	±3.69		
Rule macro + PID	394.7	5,561.5	−0.234	−0.009	0.000	26.59	5/5	5/5
	±230.8	±338.8	±0.025	±0.024	±0.000	±2.05		
Rule macro + SAC	272.6	5,533.8	−0.244	−0.019	0.000	29.95	5/5	5/5
	±32.9	±324.6	±0.006	±0.003	±0.000	±1.44		
SAC macro + SAC[†]	358.1	569.5	−0.359	−0.081	0.000	18.16	5/5	1/5
	±67.3	±4.9	±0.020	±0.020	±0.000	±1.97		

RMSE, warning fraction and age increment are lower-is-better; revenue and rewards are higher-is-better within this protocol. The age column is the simulated age-fraction increment in 10^{-4} units, not battery throughput. "Full 24h" requires all 17,280 fast ticks, 96 full macro windows and no native termination. "All gates" additionally checks service reserve, backlog, warning fraction, work conservation and two-hour recovery.

[†]**Diagnostic, not fully qualified.** All five learned-macro cases completed service and recovery, but four ended service below the 0.20 SOC reserve. Their complete-cohort means are retained; they are not recomputed from the single qualifying case. Original-bidding PPO and LLM rows are **not rerun**; old values are not inserted into this corrected panel.

What the comparison actually establishes

- **Low-level tracking improves relative to matched PID.** Retrained SAC has 30.9% lower mean RMSE and wins 4/5 paired cases. PID and SAC have identical actual market contracts in every window. The smaller SD describes these evaluation cases; it is not a claim about stability across independent training runs.
- **It is not a win on every objective.** SAC minus PID is -27.74 USD in 24-hour regulation revenue, +12.6% in service aging, and -129.69 USD in 26-hour grid cash. PID also has the higher mean native rewards. The biased peak-date Table 2 comparison is a different panel; its SAC/PID economic advantage does not automatically transfer to this nominal one.
- **The learned macro is not an improved bidding result.** It has 0/480 accepted bids: mean ask 91.90 versus mean posted price 24.94 USD/MWh. It retains the standing 5 MW commitment. The observed low revenue and reserve failures are not hidden by selecting another checkpoint after seeing this table.

These are descriptive results for fixed policies and five noise cases, not significance tests or fresh calendar-date generalization.

2. Archived Table 1, kept separate

Table 1B. Historical values only. The numbers below were extracted from `camera_ready.tex`, lines 614–648, in the original ZIP. They are not new evaluations, and their exact producing cases/checkpoint mapping were not recovered. The archived caption reports five episodes and 100% survival. No new supplemental qualification is assigned to these old rows.

Macro + hardware	RMSE (kW)	Reg. rev. (USD)	Macro reward	Low reward	Warning fraction	Age (10^{-4})
Rule macro + no agent	2,737	5,121	0.65	—	—	—
	±244	±238	±0.04			
Rule macro + Random	22,749	15	−3.86	−3.86	0.0	40.15
	±365	±17	±0.14	±0.14	±0.0	±0.07
Rule macro + Bang-bang	1,502	5,903	0.75	0.69	0.0	52.97
	±156	±344	±0.06	±0.06	±0.0	±7.03
Rule macro + Rule	621	6,092	0.94	0.87	0.0	27.23
	±90	±351	±0.02	±0.01	±0.0	±4.81
Rule macro + PID	813	6,058	0.92	0.86	0.0	27.30
	±122	±348	±0.03	±0.02	±0.0	±5.33
Rule macro + SAC	229	6,287	1.00	0.94	0.0	22.49
	±18	±353	±0.00	±0.00	±0.0	±2.78
SAC macro + SAC	780	8,550	0.94	0.85	0.0	18.38
	±626	±553	±0.13	±0.13	±0.0	±1.74
PPO macro + SAC	745	7,345	0.92	0.81	0.0	21.22
	±34	±688	±0.13	±0.17	±0.0	±2.25
LLM macro + SAC	296	2,939	0.94	0.91	0.0	20.39
	±60	±445	±0.01	±0.01	±0.0	±2.13

The same mean/±SD layout is used for readability. "—" preserves unreported archival quantities; it does not mean zero. The original paper has nine configurations, including distinct SAC, PPO and LLM macro rows. It is not a table produced by a single SAC policy. The SAC-macro RMSE SD is **626** in the requested archived source; the differing README value cannot be adjudicated without producing records.

Why the old and new columns cannot simply be subtracted

- **Reward definitions changed.** The archived paper explicitly defines throughput reward as $\alpha\theta$ and uses mean LMP times normalized facility power as its electricity-cost proxy. The old code implements those expressions. Served-dynamic-work reward and actual PCC electricity expenditure are deliberate changes of reward meaning and scale, not repairs to SAC's actor/critic update equations. Negative corrected rewards are not directly comparable to the old near-one reward values.
- **The simulated response and inputs changed.** Corrected work/idle-power handling, consistent electrical and thermal load, a complete modeled grid-import response boundary, a declared nominal reference, timestamp-faithful replay and isolated random streams change the trajectories seen by every controller. The original nominal loader advanced a five-minute source row every five seconds; matching a day label does not recreate matching inputs.
- **Evaluation and policy provenance differ.** Fast-tick RMSE is used here, aging includes the first transition, random-action streams are seeded independently, and classical controllers are fresh per case. The original Table 1 producing records are insufficient to prove which earlier metric-export path supplied each number. New SAC checkpoints were selected before this evaluation, not to maximize this new table.

Thus, comparing 229 with 272.6 kW across these panels is neither an isolated algorithm improvement nor an isolated regression. The valid current SAC/PID comparison is **within Table 1A**, where both use the same declared physical and evaluation protocol.

3. Protocol, physical checks and economic trade-offs

Service and recovery are separate. Primary columns use only 24-hour service. A fixed two-hour continuation then uses zero market revenue/commitment, full-work and nominal cooling controls, and a battery controller targeting initial SOC. Workload, weather, grid, reference and RNG states continue; the plant is not reset. All 35 cases completed recovery closure.

Table 1C. Supplemental quantities, not extra primary columns. Cash is regulation revenue minus metered electricity expenditure over the whole 26 hours. It excludes capital cost, a monetized aging charge and computing-service income; it is not total business profit.

Macro + hardware	Mean commitment (MW)	Mean grid cash 26h (USD)	Mean BESS throughput 26h (MWh)	Minimum SOC at 24h	Recovery closed	All gates
Rule macro + no agent	10.710	−54,466.21	0.000	0.5000	5/5	5/5
Rule macro + Random	10.710	−51,990.26	62.278	0.3206	5/5	5/5
Rule macro + Bang-bang	10.710	−53,605.95	86.088	0.3857	5/5	5/5
Rule macro + Rule	10.710	−53,687.22	44.351	0.4074	5/5	5/5
Rule macro + PID	10.710	−53,527.74	43.568	0.2384	5/5	5/5
Rule macro + SAC	10.710	−53,657.43	47.073	0.2963	5/5	5/5
SAC macro + SAC	5.000	−58,784.12	32.202	0.1688	5/5	1/5

Cash alone is not task performance: Random has poor tracking and very low regulation revenue despite its less-negative grid cash. This original-bidding comparison does not add the later economic maximum-capacity service floor, credit guard, full-work override or bid-quantity constraint. Lower absolute tracking error or battery use under a different commitment does not establish equal-service superiority.

The only supplemental failures. The learned macro with SAC low ends service at SOC below 0.20 in seeds 100, 101, 102 and 104. Seed 103 qualifies. All other service checks and all two-hour closures pass, including those four cases. Recovery does not erase a service-time reserve failure.

Exact primary definitions

For each five-second service tick, let G_t be actual signed grid-import power and F_t^{ref} the continuing nominal reference in MW:

$$R_t = 1000(F_t^{\mathrm{ref}} - G_t), \qquad \mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{t=1}^{N}(D_t - R_t)^2}.$$

This is not RMS of macro-window mean errors. Regulation revenue is summed using actual commitment, performance and executed duration. Electricity expenditure is $\sum_t \mathrm{LMP}_t G_t \Delta t_{\mathrm{hours}}$. Low reward is averaged over fast ticks; macro reward over returned macro windows. Warning fraction counts post-step ticks with either zone temperature at least 33°C. Aging is $10^4(a_{\mathrm{end,24h}} - a_{\mathrm{reset}})$, including the first transition.

The additional gates require full service, service-end SOC ≥ 0.20, backlog at most 1% of arrived work, warning fraction at most 0.01, work conservation and recovery closure. If any planned case were incomplete, no full-horizon primary means would be reported for that row. Qualification failures are never removed from otherwise-complete cohort means.

Exposure and configuration. Day 0 was a training start date, so this is not a fresh calendar holdout. That does not prove exact-cell or trajectory-level training exposure. Recorded training/native YAML hashes match; the Table 2 recovery YAML changes only three relative data paths to absolute paths. No static market, plant or reward-scalar mismatch was found. This is not a causal explanation of the high-level policy's rejected bids.

4. Source audit and reproducibility

The checkpoint match is explicit. Low SAC: training seed 1007, selected at 250,002 physical transitions (training continued to 2,000,016). Macro SAC: seed 1007, selected at 8,004 macro transitions. Its recorded frozen-low hash is exactly the selected low checkpoint. They have 18 observations/4 hardware actions and 19 observations/2 bidding actions, respectively. The 23-observation/1-action fractional economic controllers are different policies and were not substituted.

Low model SHA256: `5940bf534f1ff5222c5eac1554ba7a0108951df864e6a28e220b45a277a09024`
High model SHA256: `a1b5e5752d68086565c7c4fec545c80b2c702b50f07fe90ab283379898bdb388`
Registration ID: `ed5313c9b6566c816769176f0a99eac2bd9566af1f402e4cfd087437005a1971`
Pre-execution commit: `f1e15376b242851f24213fafd1a06241e8a82622`
Preserved source baseline: `34e015461a4c7e6a4fc1df8b61a72af92fc2db7a`

Independent result audit. 35/35 cases are recorded, 35 complete service and 31 pass all supplemental gates. The independent analysis passed 8,366,075 scalar checks over 105 raw trace files, checking physical tracking, both reward sums, settlement duration, price/power integration, work conservation and recovery. Exogenous/reference trajectories are paired across policies; endogenous frequency is not incorrectly treated as an exogenous fingerprint.

Verified original-code issues, versus deliberate changes

The companion `ORIGINAL_SAC_AUDIT.md` provides exact original-revision and corrected source links, numerical counterexamples and scope limits:

- **Physical inconsistencies:** command-based flex credit despite unchanged served work; idle power treated as deferrable work; heat/electrical load disagreement; incomplete response-meter boundaries.
- **Causal interfaces and clocks:** stale market offers and low-level commitments; malformed reset context; ignored custom config; accelerated source replay; shared random-stream coupling; nominal-duration settlement for an early-ended macro window. The returned RegD telemetry, not the original fast reward's observed-signal timing, was mismatched.
- **Trainer/configuration defects:** the checked-in macro SAC trainer used a rule executor rather than the claimed frozen low SAC; ignored profile/device settings; misleading checkpoint seed labels; wrong callback interfaces and a 287-tick "survival" threshold for a 17,280-tick task. This proves checked-in behavior, not an unknown historical model's producer.
- **Reporting defects:** window-mean instead of fast-tick macro RMSE, omitted first aging increment, missing no-agent fast metrics, RNG/controller state leakage, survival based on macro call count, and inconsistent export or filtering paths. Missing historical producing records limit attribution.

The throughput/cost-proxy redesign, standardized reference, wider training coverage, independent fits, early checkpoint selection and added recovery are not all "bug fixes." No one-factor ablation assigns the gains to each change. No faulty SB3 SAC update equation was established. The original simple BESS already used actual SOC-limited output.

Remaining or newly introduced limitations are not concealed. These include imperfect RegD window centering, a fixed-five-second calibration, the voltage/PCC interface and energy-equivalent workload approximation. A newer corrected macro-Box sampling-metadata defect is separately labelled: finite real-observation inference works, but generic observation-space sampling can fail. The frozen simulator and checkpoint bytes were not silently changed to improve this table.

Machine-readable evidence. `table1_values.json` and `table1_cells.csv` record every displayed metric, precision and source/model identity. `reports/table1_report.json` contains all 35 audited case summaries; `reports/raw_trace_inventory.csv` identifies the retained raw traces. The study README gives re-audit and rendering commands. No new training, LLM invocation, manuscript replacement or remote publication was performed by this study.